

MATH 1A - PRACTICE PROBLEMS FOR FINAL

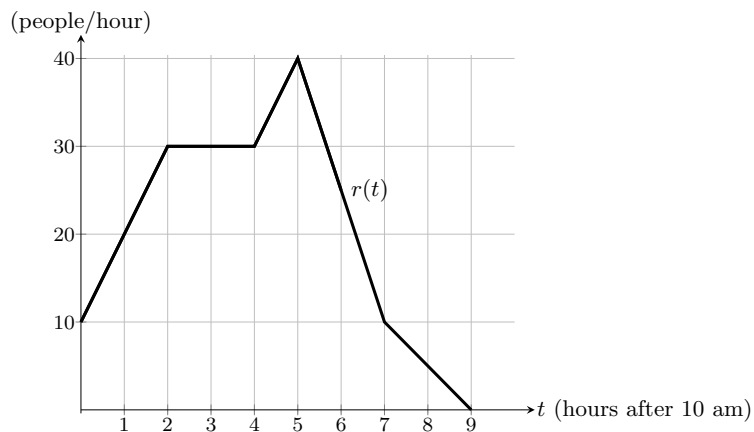
INSTRUCTIONS:

- Please print your name in the box.
- You are allowed 2 hours (120 minutes) for this exam. Pace yourself, and do not spend too much time on any one problem.
- No notes, books, calculators, computers, or other aids are allowed.
- Please **show all of your work** in the space provided and clearly indicate your answers. All of your answers should be supported by your explanation and reasoning.
- Unless otherwise indicated, please simplify your answers.
- If you need more room, use the back page and clearly indicate that you have done so. *Do not rip any pages out of this exam.*
- Please write neatly. Illegible answers will be assumed to be incorrect. **Circle or box your final answer** when relevant.

Good luck, you've got this!

Name:

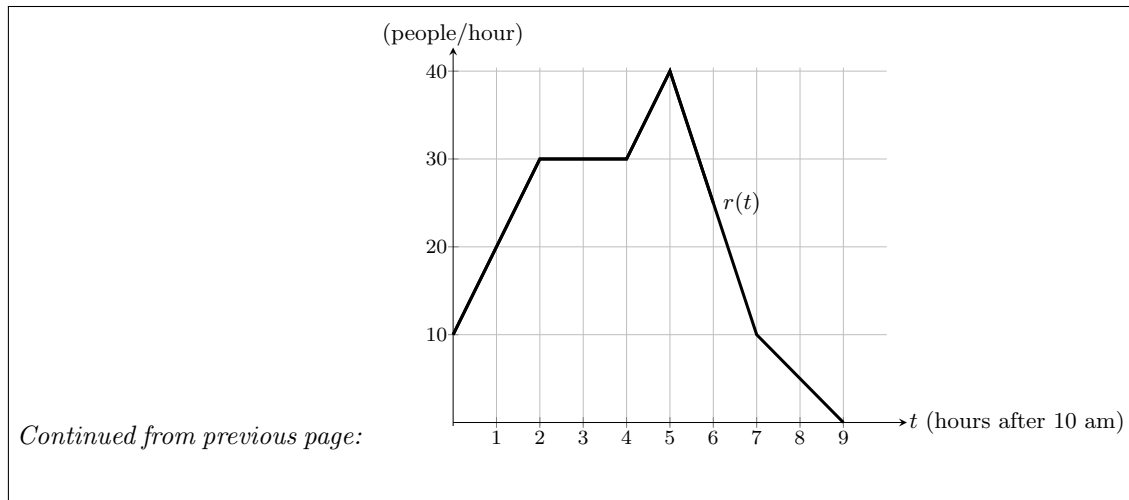
1. Your creamy avocado dip (the one that doubles as a face mask from Worksheet 16) has proved to be a giant hit - not only has the stall at the Farmer's Market taken off, you now have a brick-and-mortar store! Your store is open from 10:00 am to 7:00 pm. However, since your store is small, you can only serve 20 people per hour. Let $r(t)$ be the rate at which people arrive t hours after 10:00 am, where $r(t)$ is measured in people per hour. The graph of $r(t)$ is given to the right.



- (a) If someone arrives at 10:30 am ($t = 0.5$), how long do they have to wait to be served? Explain.

- (b) When does a line start forming?

- (c) Write and evaluate an expression involving an integral that will give you the number of people in line at 2:00 pm ($t = 4$).



- (d) If you arrived exactly at 2:00 pm ($t = 4$), how long would you have to wait in line before you got served? Explain.
- (e) At approximately what time between 10:00 am ($t = 0$) and 7:00 pm ($t = 9$) is the line the longest? Explain.
- (f) Are there still people waiting in line at 7:00 pm ($t = 9$)? How can you tell?

2. New England is a large producer of maple syrup. Several studies suggest that the amount S (in liters) of syrup produced by a typical maple tree in one year depends on that year's average winter temperature T (in $^{\circ}\text{C}$). Suppose that S and T are related by the equation

$$4T(1 + 2S)^2 = 30 + \sqrt{4S + T}$$

In 2010, the average winter temperature in Mapleton was 2°C , and the amount of syrup produced by a typical tree was 0.5 liters.

- (a) Find the value of $\frac{dS}{dT}$ then.

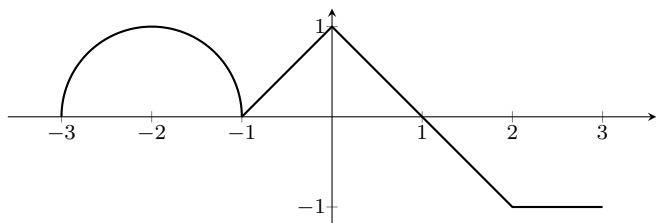
- (b) What are the units of $\frac{dS}{dT}$?

- (c) Use your answer in (a) to write a linear function approximating $S(T)$ (here we are thinking of S as a function of T) near $T = 2$. Use your linear function to estimate the amount of syrup produced per tree if the average temperature increased by $.5^\circ\text{C}$, from 2°C to 2.5°C .

- (d) Let Q be your answer to (a) $\frac{dS}{dT}$. Which of the following is the best interpretation of your answer to (a)? Circle your choice; no explanation is necessary.

- A. A typical tree in Mapleton produced about Q liters of syrup in 2011.
- B. A typical tree in Mapleton produced about $0.5 + Q$ liters of syrup in 2011.
- C. If the average winter temperature in Mapleton this year is 2.2°C , then a typical tree should produce about $0.5 + Q$ liters of syrup.
- D. If the average winter temperature in Mapleton this year is 1.8°C , then a typical tree should produce about $0.5 - 0.2Q$ liters of syrup.

3. Let f be the function whose graph is given below, and let $G(x) = \int_1^x f(t) dt$.



- (a) Evaluate the following, or say that they do not exist.

i. $G(-1)$

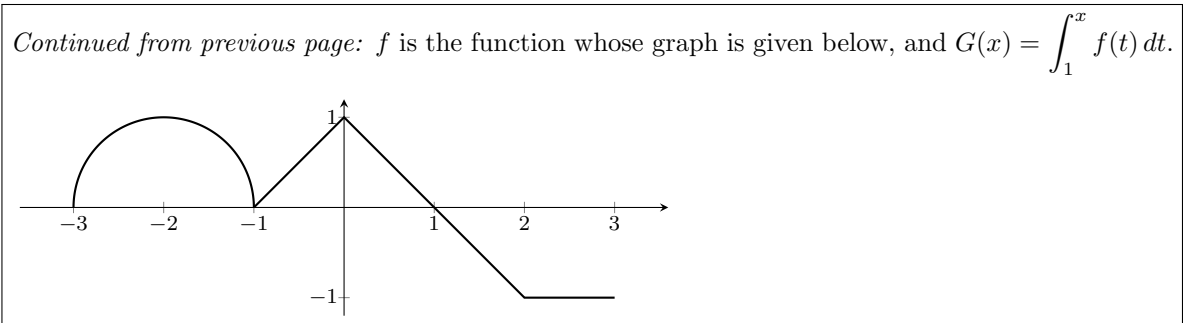
ii. $G'(-1)$

iii. $G''(1)$

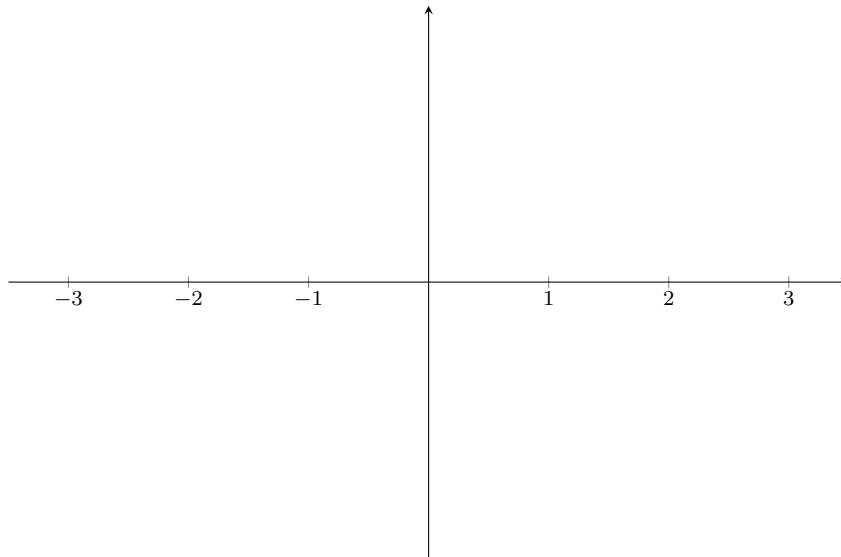
iv. $\lim_{h \rightarrow 0} \frac{G'(-1+h) - G'(-1)}{h}$

- (b) For what values of x in $(-3, 3)$ is $G(x)$ increasing?

- (c) For what values of x in $(-3, 3)$ is $G(x)$ concave up?



- (d) For what values of x in $(-3, 3)$ is $G(x)$ linear?
- (e) Sketch the graph of G on $[-3, 3]$ on the axes below with respect to information from (a)-(d).



- (f) Find all x - and y -intercepts of G on $[-3, 3]$ and label them on the graph above.

4. The Dunder Mifflin Paper Company needs to put in a design bid to get a contract account with Amazon.

The box they are designing is for Amazon's Prime Pantry service. The box must have a rectangular base whose width is twice as long as its length and has a volume of 9 cubic feet. For a competitive bid, they want to minimize the cost of each box by minimizing the surface area. What dimensions of the box should they choose to achieve that? **Make sure to justify that your answer is a minimum.**

5. Compute the following. No need to simplify.

(a) $\frac{d}{dx} (\pi \ln(1 + x^2) \sqrt{x})$

(b) $f'(s)$, where $f(s) = s^{1/s}$.

(c) $\lim_{x \rightarrow 0^+} \left(\frac{1}{x} - \frac{1}{\sin(x)} \right)$

$$(d) \lim_{x \rightarrow -\frac{\pi}{2}} \frac{\int_{-\frac{\pi}{2}}^x \sin(3t) dt}{x + \frac{\pi}{2}}$$

$$(e) \int x^2 \cdot 3^{x^3+1} dx$$

6. In preparation for the musical Hamilton, the stage crew is setting up the spotlights. When Hamilton walks left from center stage, he does so at a constant rate of 1.5 meters per second. The spotlight, located 4 meters directly above center stage follows Hamilton as he leaves the stage in a straight line path. When Hamilton is 3 meters to the left of center stage, how quickly is the angle of rotation of the spotlight (marked as θ on the sketch) changing as it follows Hamilton?

You should simplify your final answer so that it is written without using any trig or inverse trig functions.

